BT 201 Final Current

MCQS

Question and Answers

Torpor

A state of low metabolic rate and lowered body temperature, when food is scarce and night temperatures cold. Other animals can go into a state of reduced metabolism that may last several months.

How nutrient enrichment says the human is altering the aquatic and terrestrial ecosystem?

Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems.

- ***Nitrate*** concentration and export by the earth's major rivers correlate directly with human population density.
- Human disturbance also increases export of ***phosphorus*** from aquatic catchments.
- Nutrient enrichment appears to be reducing the diversity of plants and fungi in terrestrial ecosystems

Keystone species

A **keystone species** is often a dominant predator whose removal allows a prey population to explode and often decreases overall diversity. Other kinds of **keystone species** are those, such as coral or beavers that significantly alter the habitat around them and thus affect large numbers of other organisms

Role of foundation species

A **foundation species** is a **species** that has a strong **role** in structuring a community. The activities of **foundation species** physically modify the environment and produce and maintain habitats that benefit other organisms using that benefit.

Elements of landscape

A **landscape** includes the physical **elements** of geophysically defined landforms such as (ice-capped) mountains, hills, water bodies such as rivers, lakes, ponds and the sea, living **elements** of land cover including indigenous vegetation, human **elements** including different forms of land use, buildings, and structures

Foraging theory

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Factor affects the decomposition

Decomposition rate is influenced by temperature, moisture, and chemical composition of litter and the environment.

•The rate of decomposition affects the rate at which nutrients, such as nitrogen and phosphorus, are made available to primary producers. Rates of decomposition in terrestrial ecosystems are higher underarm, moist conditions.

Organelles Capable of Protein Synthesis

That is, some of the organelles within eukaryotic cells are capable of protein synthesis. These organelles with their own genetic material and protein-synthesizing capabilities are thought to have been free-living bacteria that entered into a symbiotic relationship, eventually giving rise to eukaryotic cells

Reproduction of Amoeba

Asexual reproduction

•Binary fission

Amoeba reproduces by the common asexual **reproduction** method called binary fission. ... **Amoeba** is a shapeless tiny unicellular organism that has a porous cell membrane which encloses the cell organelles and cytoplasm. **Amoeba** reproduces by the common asexual **reproduction** method called binary fission.

Importance of bacteria in environment

In summary, **bacteria** are single-celled **microbes** without a nucleus. **Bacteria** play many roles in our ecosystem. **Bacteria** are decomposers which break down dead material and recycle it. They also can be producers, making food from sunlight, such as photosynthetic **bacteria**, or chemicals, such as chemosynthetic **bacteria**.

Food web and structure of communities

Created By: Muhammad Nadeem

The feeding activities of a few keystone species may control the structure of communities

Robert Paine (1966) proposed that the feeding activities of a few species have inordinate influences on community structure. He predicted that some predators may increase species diversity by reducing the probability of competitive exclusion. Manipulative studies of predaceous species have identified many keystone species, including starfish and snails in the marine intertidal zone and fish in rivers. On land, birds exert substantial influences on communities of their arthropod prey .

Importance of protistis

Plant-like **protists** produce almost one-half of the oxygen on the planet through photosynthesis. Other **protists** decompose and recycle nutrients that **humans** need to live. ... For example, medicines made from **protists** are used in treatment of high blood pressure, digestion problems, ulcers, and arthritis.

Long 5 Marks

Examples of Symbiosis:

1. **Toxoplasma**

It is a parasitic protist that may infect a range of animals including mice, rats, and people. To reproduce sexually, the protist must infect a cat. The cat is not directly affected, but when a mouse is infected it brings harm. Mice infected with toxoplasma lose their fear of cats, which of course may bring them in close contact with a predator.

2. **Microbes**

Microbes are important for human health. For example, human breast milk contains oligosaccharides, which are short chains of sugar molecules. There is no nutritional value to babies, but the microbes are important for developing a baby's immune system.

3. **Cattle and Egrets**

Cattle do not benefit but are unaffected as egrets eat the insects that have been disturbed as the cattle forage for food.

Gene and Characteristics of Genes

Genes are functional units of heredity as they are made of DNA. The chromosome is made of DNA containing many genes. Every gene comprises of the particular set of instructions for a particular function or protein coding. Speaking in usual terms, genes are responsible for heredity

The human cell contains 23 pairs of chromosomes. The trait is one of the characteristics determined by one or more genes. Abnormal genes and genes that are formed due to new mutations also result in certain traits. Genes vary in size depending on the code or the protein they produce. All cells in the human body contain the same DNA. The difference between the cells occurs due to the different type of genes that are turned on and therefore produce a variety of proteins.

Long 10 Marks

Define Codon, nonsense codon and describe the relationship between genotype and phenotype?

CODON-A codon is a sequence of three DNA or RNA nucleotides that corresponds with a specific amino acid or stop signal during protein synthesis. DNA and RNA molecules are written in a language of four nucleotides; meanwhile, the language of proteins includes 20 amino acids. Codons provide the key that allows these two languages to be translated into each other. Each codon corresponds to a single amino acid (or stop signal), and the full set of codons is called the genetic code.

NONSENSE CODON

-There are 3 codons out of 64 in genetic code which do not encode for any Amino Acid. These are called termination codons or stop codons or nonsense codons. The stop codons are UAA, UAG, and UGA. They encode no amino acid. The ribosome pauses and falls off the mRNA. They signal release of the nascent polypeptide from the ribosome because no cognate tRNA has anticodons complementary to these stop signals, allowing a release factor to bind to the ribosome instead

Difference Between Genotype and Phenotypes

The genotype is the set of genes in our DNA which is responsible for a particular trait. Genotype is the genetic makeup of an organism and it results in some of the physical characteristics of that organism.The entire genetic information about an organism is contained in a genotype.. Genotype is determined by the makeup of something called "alleles

The phenotype is the physical expression, or characteristics, of that trait. For example, two organisms that have even the minutest difference in their genes are said to have different genotypes. These two mice may have different genotypes and both can still be white if they share the particular phenotype for white fur.

The phenotype is the physical appearance of an organism, while the Genotype is the genetic composition of an organism. Phenotype is observable and is the expression of the genes of an individual. So even the organism with the same species may differ, with a minute difference in their genotype. This is the main difference between the two.

We can notice one's hair colour, eye colour, height, weight, skin colour, etc. but cannot look at genes responsible for these characters, so the observable physical look is the phenotype while the unnoticed genes responsible for such characters present in the DNA of cell of the individual is genotype. (From ppts)

Populations include genetic and phenotypic variation among individuals, which are significant aspects of population structure. The first biologists to conduct thorough studies of phenotypic and genotypic variation and to incorporate experiments in their studies, focused on plants. By using plants cloned from the same parent, Bonnier was able to control for the effects of genotype on plant form and observe the effects of climate.